

El Mehdi Rahali

(Software Engineer, C#/.NET)

✉ mrahali.de
📍 Siegen, Germany

🔗 github.com/printfmehdi
🌐 linkedin.com/in/el-mehdi-rahali

☎ +49 151/53305475
✉ mehdi1rahali@gmail.com

Summary

Software Engineer with 3+ years of backend experience in C# and .NET. Specialised in scalable APIs, distributed systems, and Clean Architecture. Experienced in demanding enterprise environments with high requirements for stability, multi-tenancy, and production-grade reliability.

Skills

Architecture & Systems: Clean Architecture, DDD, Distributed Systems, Event-Driven Architecture

Backend & Frameworks: ASP.NET Core, Entity Framework Core, LINQ, Dapper

APIs & Integration: REST APIs, Webhooks, OpenAPI/Swagger, OData, Gridify, OAuth2, Kiota.

Workflow & Automation: Temporal, Quartz, N8N

Cloud & DevOps: Azure, Docker, Podman, Kubernetes, CI/CD

Databases & Search: PostgreSQL, Microsoft SQL Server, MongoDB, Elasticsearch, MinIO (S3)

Observability & Tooling: Serilog, Grafana, Elastic Stack, Git, Jira

Languages: C#, .NET, Java, Python, JavaScript

Experience

Backend Engineer — Distributed Systems (.NET / C#)

Aug 2024 - Present

TKD GmbH — Germany

- Designed and implemented a modular monolith backend architecture for a multi-tenant telecommunications platform, managing devices, contracts, tariffs, and billing workflows across enterprise customers
- Built event-driven architectures using webhooks and orchestrated workflows with Temporal, enabling reliable processing of external events (e.g., Provisioning of new Tenants, shipment tracking, contract updates) and improving system responsiveness
- Integrated multiple external enterprise systems (e.g., Entra, Telekom B2B, ServiceNow, WeClapp, Zendesk) to automate contract, tariff, and service data synchronization, ensuring consistency across distributed systems under continuous data flow
- Architected and implemented an importer/exporter service hub to process and synchronize large volumes of external data (devices, contracts, services), improving data consistency and reducing manual operations
- Contributed to the architecture and implementation of a multi-tenant shop system (TKD Shop), including product modeling, pricing, and cross-selling features for customer-facing use cases
- Enhanced system performance and reliability through distributed caching, structured logging (Serilog), and monitoring with Datadog

Software Developer (Working Student)

Nov 2021 - Jun 2024

Buhl Data Service GmbH — Germany

- Contributed to the backend development of a document management System, orchestrating the full lifecycle of tax documents, including upload, antivirus scanning, OCR processing, metadata extraction, and document versioning.
- Worked with Alfresco-based storage for audit-compliant document management, retrieval, and version control.
- Integrated Elasticsearch as a search engine for efficient indexing, querying, and enabling full-text search across large document datasets.
- Integrations with multiple consumer channels, including customer-facing apps, German tax authorities, and other internal BI systems.
- Monitored and optimized production systems using Grafana and Elastic Stack, improving observability and reducing incident response time.
- Development and maintenance of CI/CD pipelines in Azure DevOps to automate build and release processes.

Projects

Bachelor Thesis — Feature Model Learning Optimization

- Optimized feature model learning using dimensionality reduction (Linear PCA, Logistic PCA)
- Conducted empirical evaluation based on precision, recall, and runtime
- Built automated experimentation pipelines for model training and evaluation

Tech: Python, R, AutoML, AutoGluon, Shell Script, Singularity, Git, LaTeX

Restaurant SaaS

- Built a complete end-to-end restaurant platform, featuring real-time communication, a user-friendly ordering experience with a dynamic UI, customizable product options, and an admin dashboard for restaurant management
- Implemented real-time order tracking using SignalR, enabling live synchronization between kitchen systems and customer-facing UI
- Developed REST APIs using ASP.NET Core and implemented CQRS with MediatR to ensure clear separation of concerns and maintainable request handling
- Applied Clean Architecture principles to enforce modularity, testability, and long-term scalability of the system
- Integrated PostgreSQL for reliable data persistence and optimized query performance for order and product management

Tech: .NET, React, SignalR, TypeScript

9raytiDrive (EdTech) — Ongoing

- Developing an MVP of a document-sharing platform for students to upload, discover, and organize academic resources
- Designing the backend architecture using ASP.NET Core, Clean Architecture, CQRS, and PostgreSQL
- Implementing authentication, authorization, and document management features to support the platform's core workflows
- Building the foundation for scalable search, file storage, and multi-tenant capabilities

Tech: .NET, ASP.NET Core, PostgreSQL, Clean Architecture, CQRS, Identity, JWT, React, TypeScript

AI Video Automation System — Ongoing

- Built an AI video generation system automating the pipeline from script generation to final output
- Designed LLM-based agents for content generation, TTS, and media composition workflows
- Improved reliability by stabilizing API integrations and handling asynchronous failures

Tech: Python, LLM APIs, AI Agents, TTS Systems, FFmpeg

Education

B.Sc. Computer Science

Sep 2020 - March 2024

University of Siegen, Germany

Specialisation: Software Engineering

Preparatory Cycle in Engineering

Sep 2018 - Feb 2020

National School of Applied Sciences, Morocco

Languages

German (C1), English, French, Arabic (Native)